Lab 5: Before and After Test Results

Program: pharmacy_system.py (Pharmacy Prescription & Stock Tracker)

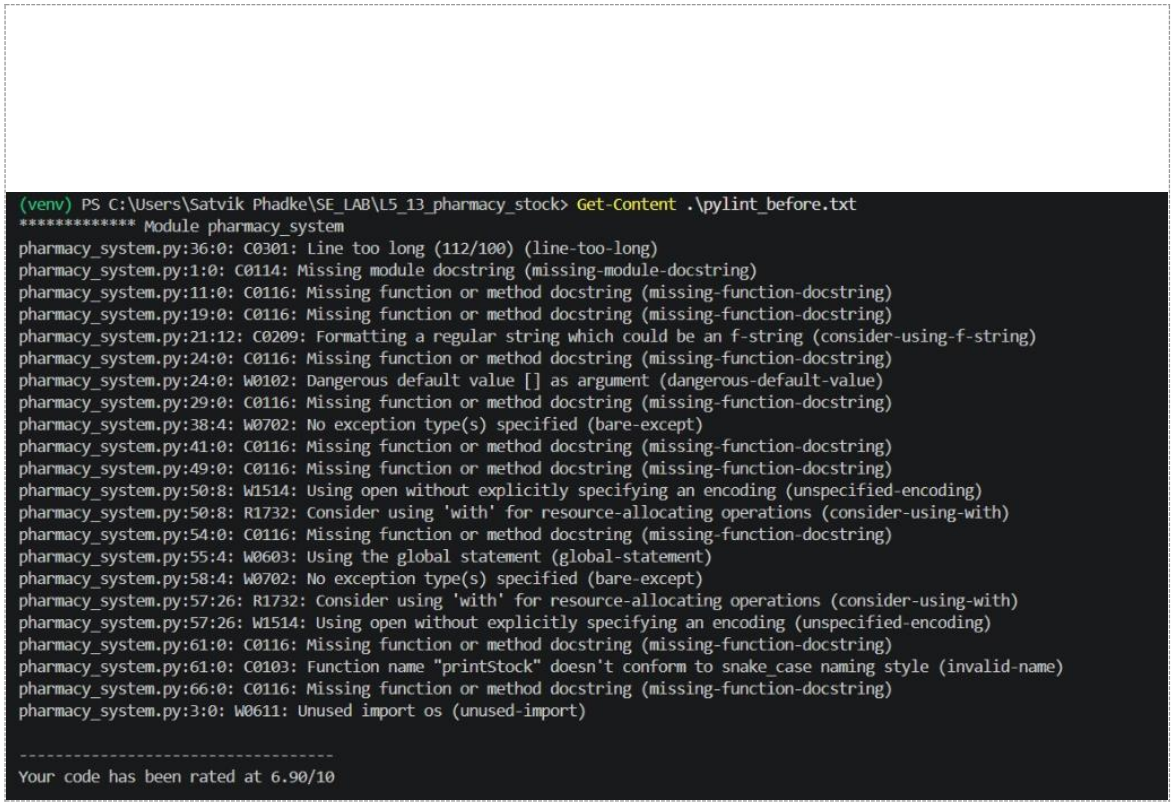
NAME : Satvik Phadke SRN : PES2UG24CS450

Summary

Tool	Before fixes	After fixes
Pylint	6.90/10	10.00/10
Flake8	15 warnings	0 warnings
Bandit	2 issues (1 Medium, 1 Low)	No issues identified

Pylint

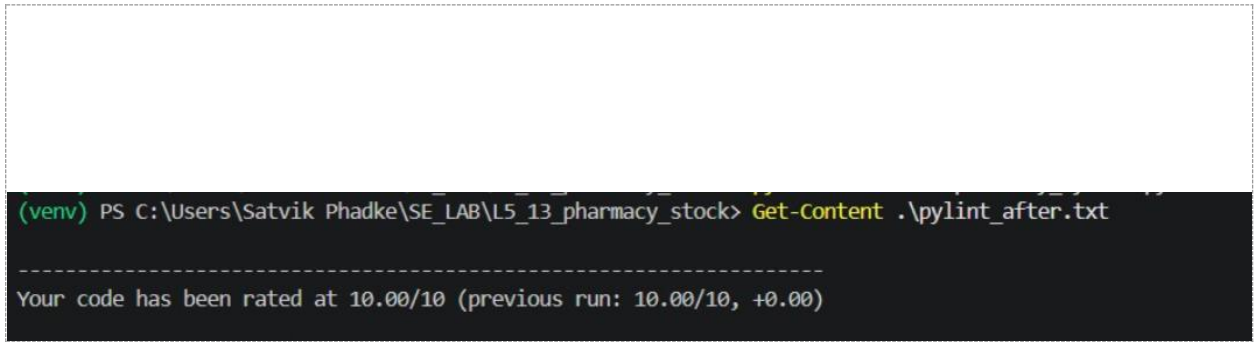
Before fixes (6.90/10)



```
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\pylint_before.txt
************* Module pharmacy_system
pharmacy_system.py:36:0: C0301: Line too long (112/100) (line-too-long)
pharmacy_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
pharmacy_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:19:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:21:12: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
pharmacy_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:24:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
pharmacy_system.py:29:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:38:4: W0702: No exception type(s) specified (bare-except)
pharmacy_system.py:41:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:49:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:50:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
pharmacy_system.py:50:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
pharmacy_system.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:55:4: W0603: Using the global statement (global-statement)
pharmacy_system.py:58:4: W0702: No exception type(s) specified (bare-except)
pharmacy_system.py:57:26: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
pharmacy_system.py:57:26: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
pharmacy_system.py:61:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:61:0: C0103: Function name "printStock" doesn't conform to snake_case naming style (invalid-name)
pharmacy_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
pharmacy_system.py:3:0: W0611: Unused import os (unused-import)

-----------------------------------
Your code has been rated at 6.90/10
```

After fixes (10.00/10)

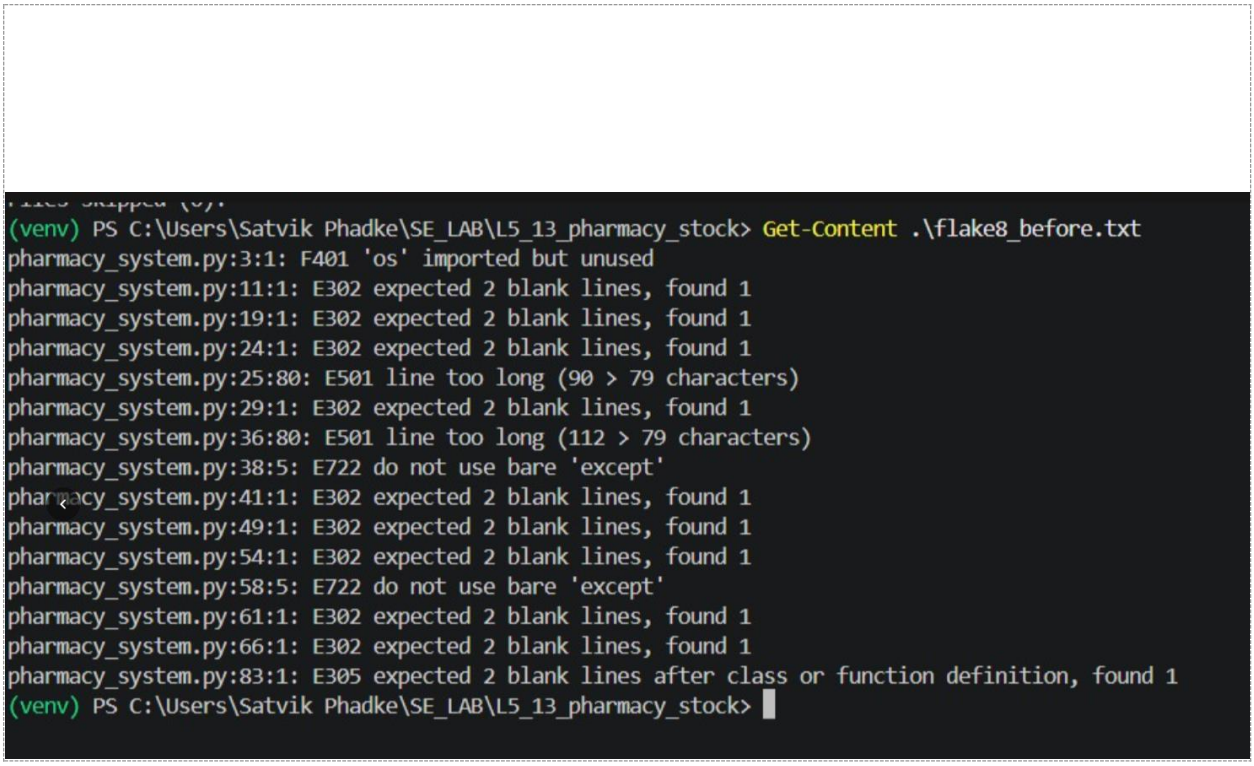


```
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\pylint_after.txt

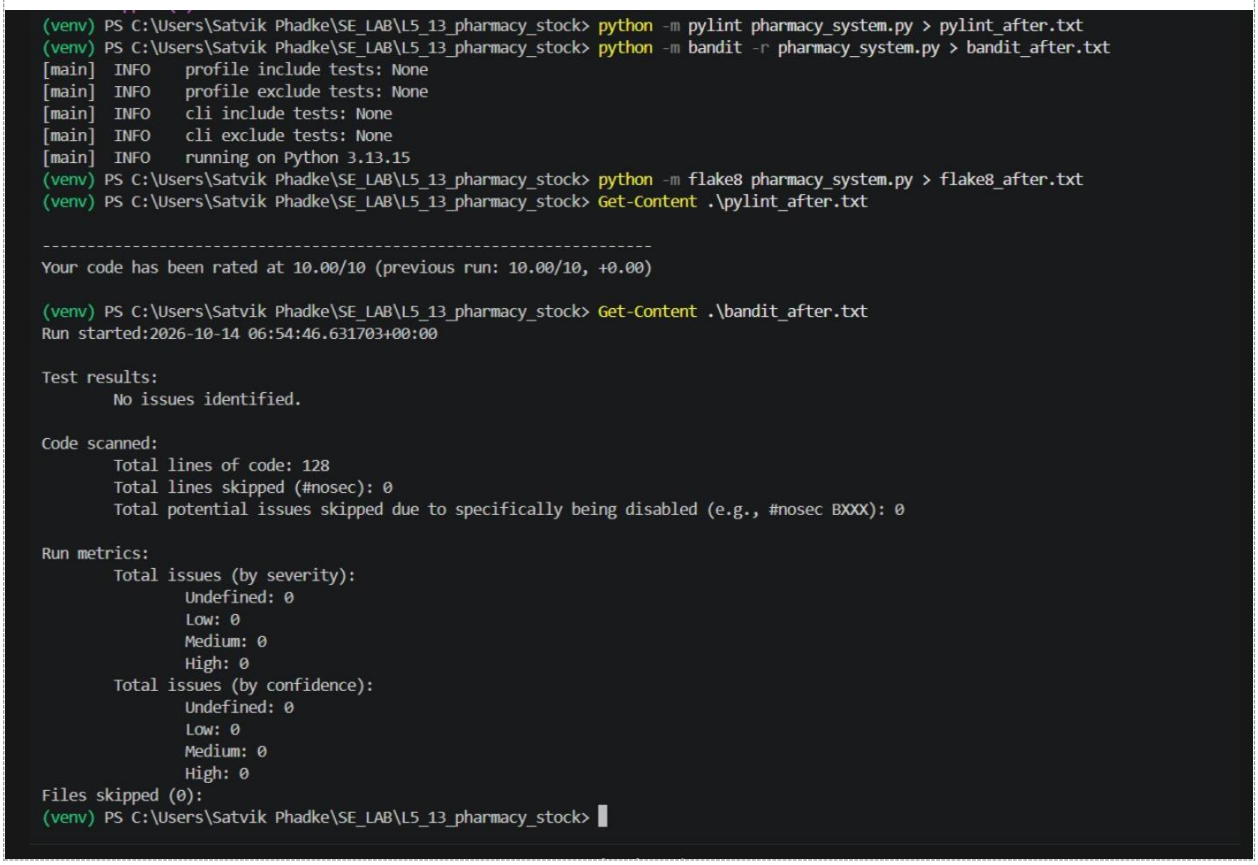
-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Flake8

Before fixes (15 warnings)



After fixes (0 warnings)



```
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> python -m pylint pharmacy_system.py > pylint_after.txt
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> python -m bandit -r pharmacy_system.py > bandit_after.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.15
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> python -m flake8 pharmacy_system.py > flake8_after.txt
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\pylint_after.txt

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\bandit_after.txt
Run started:2026-10-14 06:54:46.631703+00:00

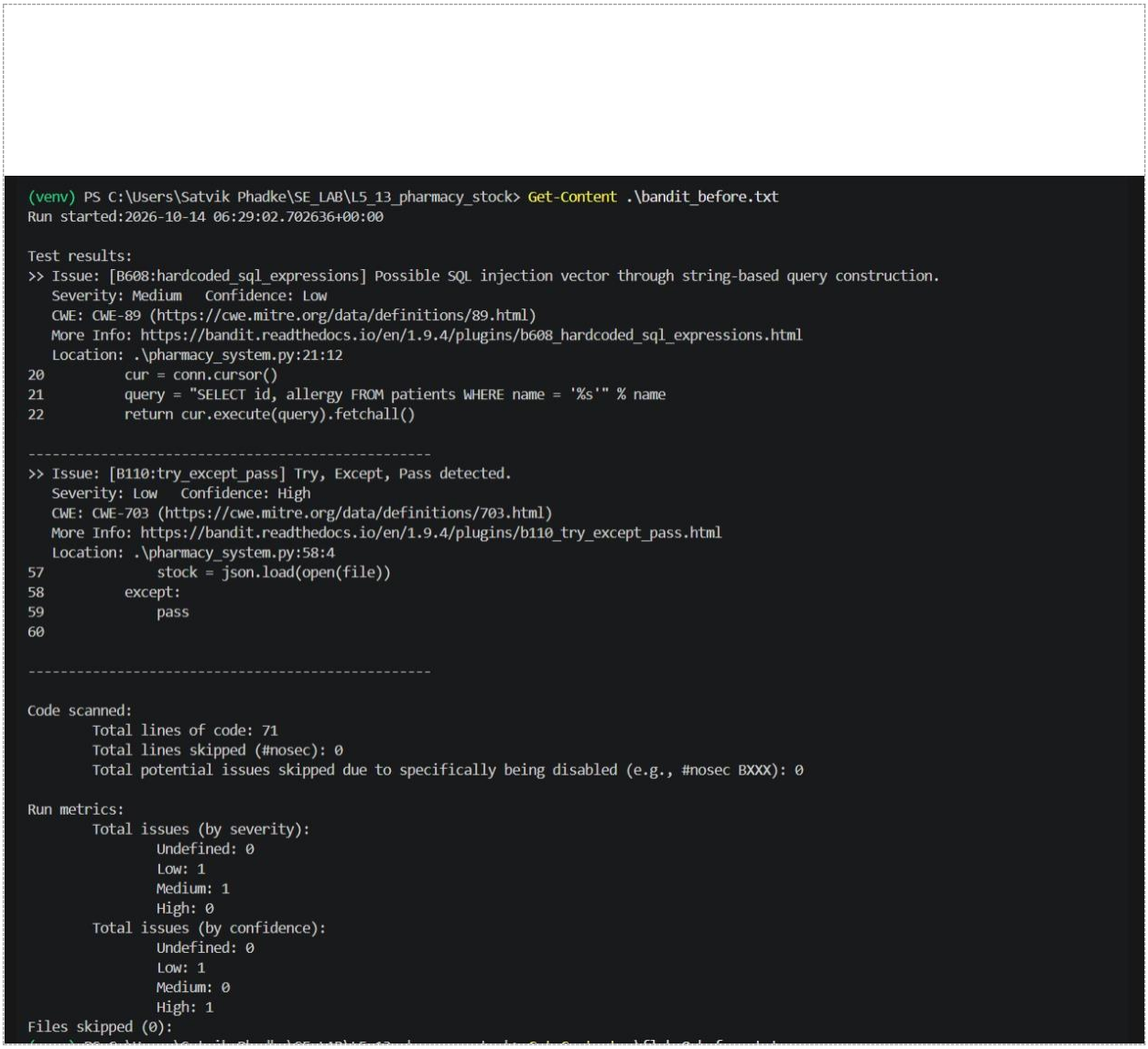
Test results:
        No issues identified.

Code scanned:
        Total lines of code: 128
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock>
```

Bandit

Before fixes (2 issues)



```
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\bandit_before.txt
Run started:2026-10-14 06:29:02.702636+00:00

Test results:
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Low
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: .\pharmacy_system.py:21:12
20          cur = conn.cursor()
21          query = "SELECT id, allergy FROM patients WHERE name = '%s'" % name
22          return cur.execute(query).fetchall()

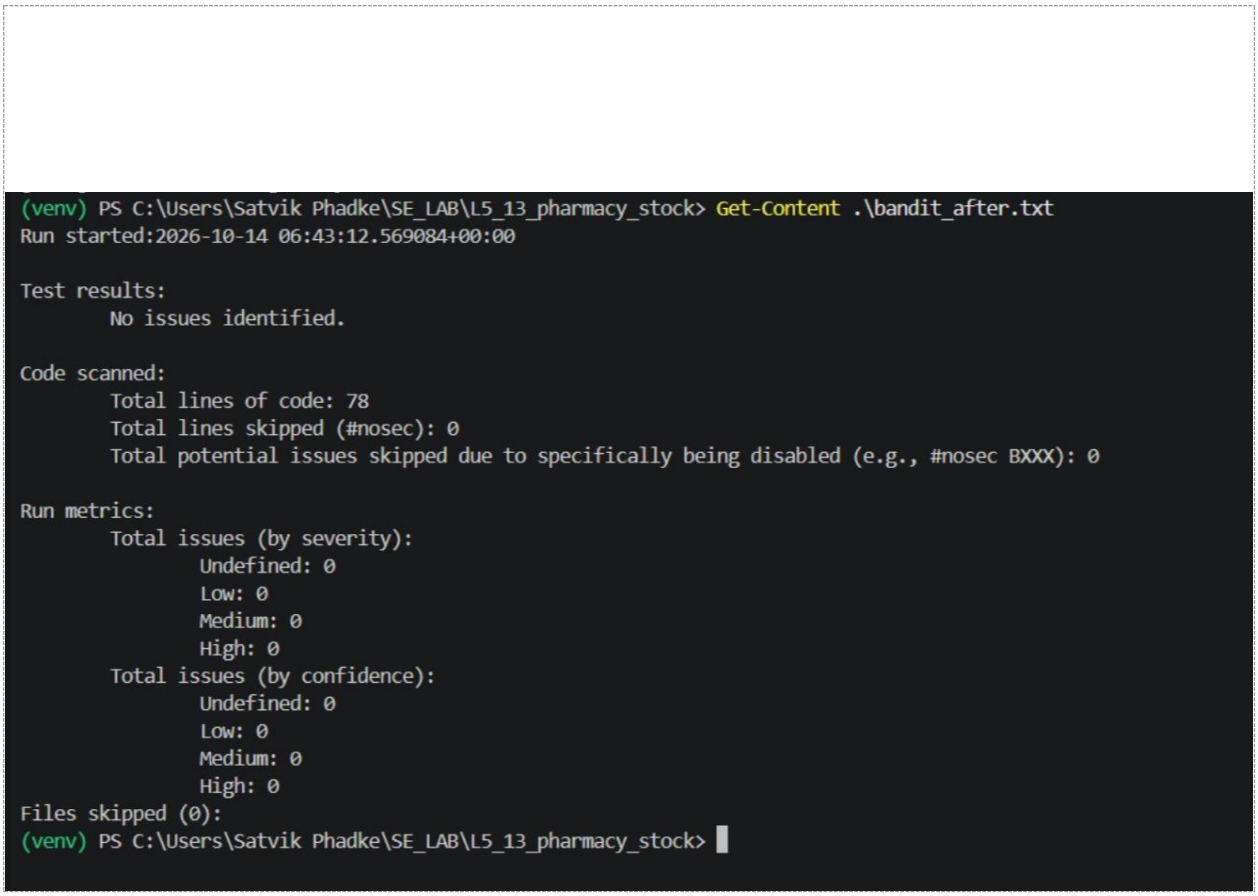
--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\pharmacy_system.py:58:4
57              stock = json.load(open(file))
58          except:
59              pass
60

--------------------------------------------------

Code scanned:
        Total lines of code: 71
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 1
                Medium: 0
                High: 1
Files skipped (0):
```

After fixes (no issues)


(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock> Get-Content .\bandit_after.txt
Run started:2026-10-14 06:43:12.569084+00:00
Test results:
No issues identified.
Code scanned:
Total lines of code: 78
Total lines skipped (#nosec): 0
Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0
Run metrics:
Total issues (by severity):
Undefined: 0
Low: 0
Medium: 0
High: 0
Total issues (by confidence):
Undefined: 0
Low: 0
Medium: 0
High: 0
Files skipped (0):
(venv) PS C:\Users\Satvik Phadke\SE_LAB\L5_13_pharmacy_stock>